Toronto Metropolitan University

Department of Electrical, Computer & Biomedical Engineering
Faculty of Engineering & Architectural Science

Course Title:	**Fundamentals of Data Engineering**
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Instructor:	**Dr. Yasaman Ahmadiadli**
Lab Report Number	**Lab 2**
Lab Title	**Lab 2: Entity-Relation Diagram Design**

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Entities:
Restaurant: A physical restaurant establishment, that accepts reservations and serves customers food
Customer: The client/guest who makes the reservation and places orders on the restaurant
Table: The seating unit within the restaurant
Reservation: A booking made by a specific customer for a certain time with a table preference
Order: The actual purchase order made by the customer through reservation or even a walk-in
MenuItems: The list of food(or drinks) and items offered by the restaurant
Staff: Employees operating the restaurant

Relationships:
Restaurant - *accept* - Reservation: Restaurant acknowledges reservations and keeps the details
Restaurant - *has* - Table: Restaurant possesses tables as seating units for the establishment
Restaurant - *confirm* - Order: Restaurant verifies the order, whether it be reservation or walk-in
Restaurant - *Employ* - Staff: Restaurant hires employees for operation
Reservation - *assigns* - Table: Reservation allocates preferred customer table to details and thereby books table
Reservation - *generate* - Order: The Reservation customer make their order accordingly
Order - *contain* - MenuItems: The order contains some of the following MenuItems provided
Staff - *handle* - Order: The Staff handle the following orders being made by the customers
Staff - *arrange* - MenuItems: The Staff are in charge of choosing and deciding the items for the menu
Customer - *make* - Reservation: the Customer makes the reservation accordingly
Customer - *make* - Order: The customer also has the choice of just walking-in and ordering directly

Attributes:
Restaurant:-<u>restaurant_ID:</u> Unique attribute representing a ID
-address: The location of the restaurant
-name: Restaurant name
-phone_number: Phone number of the restaurant
-opening_Hours: multivalued as there might be different hours for each day of the week
-cuisine_type: type of food and style for customer information
-total_capacity: capacity of the restaurant
Customer: -<u>customer_ID:</u> Unique attribute representing a ID
-first_name: first name of customer
-last_name: last name of customer
-phone_number: Phone number of the customer
-email: email address of customer
-loyalty_status?: Is the customer a loyal member of the restaurant or not
Table: -<u>table_ID:</u> Unique attribute representing a ID
-table_number: number assigned to the table
-seating_capacity: capacity for the table itself
-availability_status: Is it available to book or not
-seating_location_description: multivalued attribute to describe how it looks around the table(Window, patio…)
Reservation: -<u>reservation_ID:</u> Unique attribute representing a ID
-reservation_date: date for reservation
-reservation_time: time for the reservation
-reservation_status: Is the specific reservation for time, date, and table available?
-number_of_guests: guests amount

Order: -order_ID: Unique attribute representing a ID
-order_date: date for specific order
-order_time: time for the order made
-order_status: Is the specific order(with the meuItems) available?

Order: -menu_item_ID: Unique attribute representing a ID
-item_name: name of the item
-price: price for the item
-availability_status: Is the item still available?

Staff: -staff_ID: Unique attribute representing a ID
-role: staff positions, etc. cashier, cook, cleaner, manager…
-first_name: first name of employee
-last_name: last name of employee
-hourly_wage: For company and database purposes, not shown to clients or non-owners

Relation *“assigns”*: -seating_startTime: Start time for the reservation guests to occupy their table
-seating_endTime: End time for the reservation guests to occupy their table